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Exploring demand and supply-side barriers to safe water, sanitation, and hygiene services in urban slums in southeast Nigeria

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Abstract

Background Optimal access to safe water, sanitation, and hygiene (WASH) services remains a significant challenge in urban slums. Understanding the demand- and supply-side barriers to the provision of safe WASH facilities and the factors influencing access to these services will contribute to developing effective strategies for promoting such services and improving the health of slum dwellers. This paper provides new evidence on demand- and supply-side barriers to safe WASH services in urban slums in Nigeria.

Methods A qualitative study was undertaken in two urban slums in Okpoko, Ogbaru LGA, Anambra state, Southeast, Nigeria. In-depth interviews were conducted with 22 WASH stakeholders, and six focus group discussions were held with community members. Data were analyzed thematically using NVivo software.

Results Barriers to the adoption of safe WASH practices and services exist at multiple levels. On the demand-side, intrapersonal-level factors, such as limited knowledge, attitudes, and practices of safe WASH and its benefits, hinder access to WASH services. Additionally, interpersonal-level factors, including a lack of demand creation for WASH services, also pose a challenge. Conversely, supply-side factors that impede the provision of WASH infrastructure and services include community-level barriers, such as weak leadership and governance, institutional-level barriers, including inadequate multisectoral collaboration, inspection, and implementation of WASH regulations, and public policy-level barriers, such as insufficient funding and resource allocation at state and local government level.

Conclusion The study identifies a range of barriers affecting access to and provision of WASH services in urban slums, including individual, institutional, and policy-level constraints. To address these challenges, policy recommendations include integrating WASH education, strengthening multisectoral collaboration, enhancing institutional capacity, and increasing funding and resource allocation for WASH infrastructure.

Keywords Barriers, WASH, Nigeria, Supply- and Demand-side, Urban Slums



1 Introduction

About 2.3 billion individuals globally lack access to basic sanitation facilities, and 844 million lack access to basic drinking water services [1]. Access to safe and affordable water, sanitation, and hygiene (WASH) services is a fundamental human right, essential for public health, dignity, socioeconomic development, and overall well-being of the people [2]. Hence, the Sustainable Development Goal 6 (SDG 6) aims to ensure universal access to safe WASH by 2030. However, urban slums in low-and middle-income countries (LMICs) continue to face significant challenges in accessing these basic services [1].

Nigeria's WASH indices have persistently remained below global targets with notable regional and socioeconomic disparity. For instance, only 20 million individuals, equivalent to 10% of the Nigerian population, have access to complete basic WASH services, while 67% lack access to safe drinking water supply services [3]. In addition, only 46% of the population were found to have access to basic sanitation services, 23% still defecate in the open, and only 17% of household members have access to basic hygiene services [3]. Notably, Nigeria overtook India in 2019 to become the number one open defecation nation globally [4], and approximately 50 million Nigerians (or 10 million households) defecate in the open [5]. Evidence shows that the lack of access to WASH services was very poor in urban slums, as 73% of urban slum dwellers lack access to improved sanitation facilities [5], and 64% lack access to safe drinking water [6]. Furthermore, urban slums have higher rates of water-borne diseases, such as diarrhoea and cholera [7].

The poor access to basic WASH facilities in Anambra State, southeast Nigeria, mirrors these national trends [8]. Similarly, reports show that urban slums contribute significantly to the burden of WASH poor outcomes in Anambra state. For instance, 71% of households in urban slums lack access to improved sanitation facilities, and 55% lack safe drinking water [9, 10]. In addition, poor WASH conditions in urban slums were found to contribute to the high prevalence of water-borne diseases [11].

Improving access to WASH services will reduce health-related risks, years spent schooling, and diseases [12–14]. Promoting the provision and use of safe WASH services is a cost-effective strategy and a useful public health measure for infectious disease prevention and control [15, 16].

However, urban slum dwellers face several challenges in accessing and providing WASH services, especially since urbanization and population growth have increased the demand for WASH services and available resources and infrastructure are inadequate to meet this demand [17]. These challenges include both supply-side and demand-side barriers. Several studies have identified demand-side barriers to access to WASH, including socioeconomic factors, cultural and behavioural practices, and poor awareness and knowledge of the importance of safe WASH practices that deter access to WASH services [5, 18, 19]. On the other hand, supply-side barriers that limit the provision of WASH services were reported to include inadequate facilities or infrastructure [18], insufficient institutional capacity (e.g., governance and regulation) [20]; and poor funding/financial constraints [1].

In Nigeria, urban slum dwellers have limited access to basic services, including WASH [21] and high prevalence of water-borne diseases, e.g. diarrhoea and cholera [11]. These occur despite the recognized importance of improving WASH access in achieving sustainable public health outcomes.

There is a paucity of evidence on the demand and supply-side barriers to WASH services in urban slums in Nigeria. A systematic understanding of these barriers will provide a comprehensive understanding of how multi-level factors interact to affect WASH access. Hence, the paper provides new knowledge on demand and supply-side barriers to access to WASH services in urban slums in southeast Nigeria. Overall, the paper contributes to the growing knowledge of WASH in low-income urban settings and can inform policy and programmatic decisions for improving access to these essential services. The findings also provide the evidence-based basis for the development of targeted interventions and policies to enhance WASH services in urban slums and thereby contribute to progress toward SDG 6 and improved public health outcomes in Nigeria.

2 Methods

2.1 Conceptual framework

The study employed a Demand-Supply framework [22] and the Socio-Ecological Model (SEM) (Fig. 1) [23] to gain a deeper understanding of the challenges or barriers to WASH. The framework categorized the barriers into demand-side barriers, which impact households' access to WASH, and supply-side barriers, which affect the availability and quality of WASH services.

The SEM adds a layered perspective by examining how these barriers operate at multiple levels - individual or intrapersonal level with factors like socioeconomic status and awareness of WASH-related health benefits. The second level is the household or interpersonal level, which involves family/household practices, including cultural or behavioural practices that may hinder safe WASH. The third level is the institutional or organizational level, which looks at the organizations within the community whose efforts impede the uptake of WASH interventions. The next level, the community level, concerns local infrastructure, resources, and the community's ability to demand and

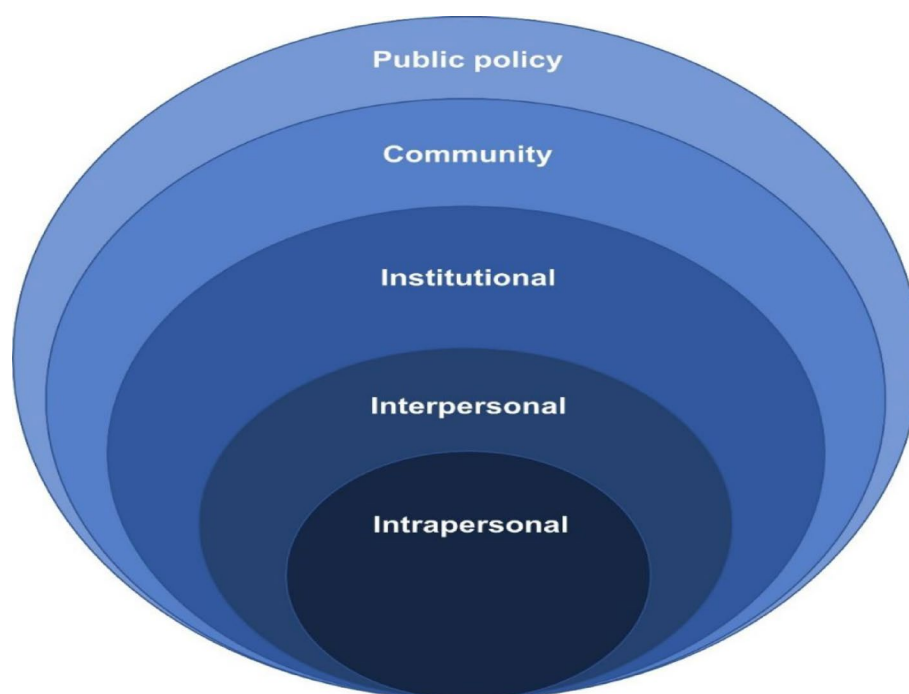


Fig. 1 Overview of the SEM proposed by Mcleroy and Colleagues Mcleroy et al. [23]

sustain WASH services. The last level is the political or policy level, focusing on governance and/or funding that hinder the provision of WASH services.

2.2 Study design and area

A qualitative study was conducted in two communities in Okpoko slums, Anambra state, southeast, Nigeria. The 2022 population estimate of Anambra is 6.1 million. Okpoko is the largest and most populous urban slum in Anambra state. It sits on the east bank of the Niger River just north of the larger city of Onitsha. Okpoko has been identified as a classic slum characterized by haphazard development, very high-density living and a lack of public amenities and facilities, including WASH [17].

2.3 Study population and selection of participants

The study population comprised policymakers, program managers, community leaders, health service providers and community members. Participants were purposively selected either based on their involvement in WASH programming at state and local government area (LGA) levels in Anambra State and/or based on their roles as leaders or decision-makers in the community. The purposive sampling was done to achieve representation of various groups and diverse experiences. Specifically, the participants were recruited from (i) State government ministries and agencies, including the Ministry of Health, Ministry of Environment, Ministry of Information, Ministry of Power and Water Resources, Ministry of Education; (ii) Ogbaru LGA Health and WASH departments; (iii) Health Providers; (iv) Community leaders including President General, Chief Security Officer, Youth Leader, and Women Leader and; (v) Community members (men, women and adolescents). The participants were identified before data collection through a stakeholder engagement workshop organized by the researchers to map key stakeholders of WASH at all levels and to determine the study population for the qualitative interviews.

2.4 Data collection

Pretested semi-structured IDI and FGD guides were the data collection tools. The IDI participants were interviewed at convenient locations for respondents, such as at the health facilities and their offices. At the same time, the FGDs were conducted in health facilities that were convenient places for the participants. Interviews were conducted in English or Igbo language, according to respondents' preferences and transcribed verbatim, then translated from Igbo language into English. The average time spent on the IDI and FGD were 40 min and 60 min, respectively. The data was collected between March and April 2024. A total of 22 in-depth interviews (IDIs) and six sex-disaggregated focus group discussions (FGDs) were conducted. The characteristics of the IDI and FGD participants are summarized in Table 1.

Before conducting the IDIs and FGDs, all participants provided written and verbal consents after being informed of the purpose of the study, the benefits and risks of participating in the study, their rights to voluntary participation and the confidentiality of data. In addition, permission to audio-record interviews/discussions was also obtained.

2.5 Data analysis

Thematic analysis was conducted using NVivo software to systematically organize and interpret the qualitative data. All audio files were transcribed and translated where

Table 1 Socio-demographic characteristics of the participants

Variables	IDI Respondents-Policy/decision-makers and leaders (N=22)		FGD participants-Community members (N=40)	
	Variable	n	Variable	n
Category/level of operation	State-level policymakers/program managers	10	Men group	11
	LGA-level policymakers/program managers	5	Women group	14
	Community-level policymaker/provider	7	Adolescent group	15
Age (years)	–	–	Less than 20 years	9
	–	–	20–30	13
	31–40	4	31–40	5
	41–50	7	41–50	4
	51–60	6	51–60	3
	60 and above	5	60 and above	2
Sex	Male	11	Male	14
	Female	11	Female	26
Highest level of education	Primary	0	Primary	2
	Secondary	5	Secondary	34
	Tertiary and above	17	Tertiary	4
Position/designation	Managing director	3	Unemployed	9
	Program managers/coordinator/HOD	5	Artisan	4
	Desk officer/program officer	7	Employed in private or public	10
	Community leader	5	Trader	10
	Health provider	2	Businessman/woman	3
			Others (Native doctor, security agent)	4

necessary. Then, the transcripts were validated against field notes for accuracy and completeness. Each transcript was subjected to multiple readings by two researchers to identify text patterns. Discrepancies from the outputs were resolved through double review, ensuring inter-code reliability. The coding identified similar/related responses, which were grouped into themes and categories used to interpret and report the findings.

Data was coded and categorized using the Demand-Supply Framework integrated with the five levels of SEM to capture the barriers affecting access to WASH services. The demand-side barriers were analyzed by categorizing factors influencing individuals' ability to seek and adopt WASH services across the SEM levels. A cross-level analysis was conducted to explore how demand and supply-side barriers interact across SEM levels. For instance, whether the policy-level constraints in funding and regulation (supply-side) directly influenced the community-level availability of infrastructure, which subsequently impacted the household-level practices and individual-level awareness, shaping overall WASH accessibility and adoption. This approach allowed for a comprehensive understanding of the complexity of multi-level barriers to the provision of and access to safe WASH access. The methodological approach to the data analysis is illustrated in Fig. 2.

3 Results

The findings highlight the interplay between demand and supply-side barriers to WASH access in urban slums. The results are presented subsequently according to themes described in the data analysis section. Table 2 shows a summary of barriers to access and provision of WASH services.

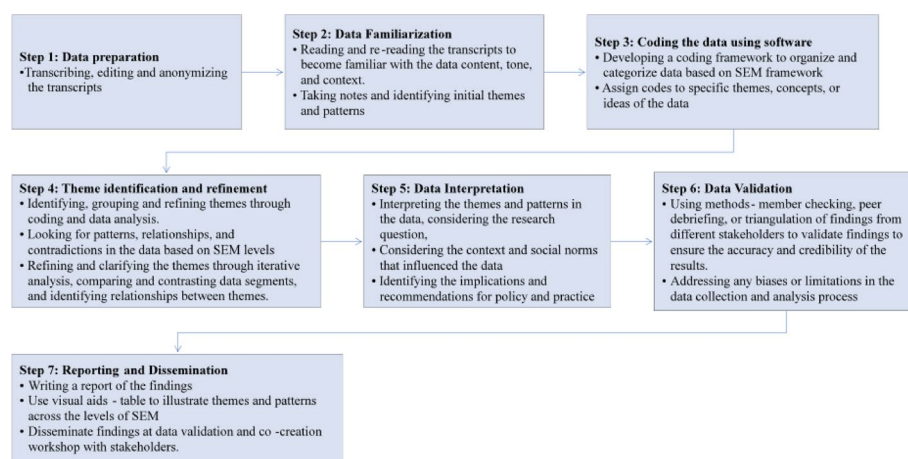


Fig. 2 Methodological approach to the data analysis

Table 2 Summary of barriers affecting access to and provision of WASH services in urban slums

Demand-side barriers	Supply-side barriers
Intrapersonal level • Limited knowledge, attitude and practice of safe WASH practices and its benefits • Limited awareness and understanding of WASH benefits • Hostility of the inhabitants/ individuals towards refuse service providers and community sanitation task force Interpersonal-level • Lack of demand creation for improved or safe WASH practices • Refusal/lack of funds to pay sanitation levy • Affordability and cost of WASH services particularly water	Institutional-level • Inadequate multisectoral collaboration in WASH • Inadequate inspection of WASH facilities by the relevant authorities, including - Infrequent inspections - Lack of technical expertise for inspection and - Limited enforcement of standards • Lack of maintenance and repair of existing public boreholes • Inadequate implementation of rules/laws against open defecation and indiscriminate refuse dumping • Inadequate waste management services/ facilities - Irregular evacuation of refuse by the public service providers - Insufficient waste disposal infrastructure Community-level • Weak community leadership and governance in the enforcement of WASH rules, including: - Lack of accountability mechanisms for community leaders - Limited capacity of community leaders to mobilize residents for WASH programs and services - Poor enforcement of community-level WASH regulations - Inability to safeguard and protect public boreholes against vandalization of their parts • Inadequate water supply system to serve the population, including: - Insufficient water infrastructure to meet the needs of the growing population - Poor maintenance of existing water supply systems - Irregular or unreliable water distribution • Absence of and insufficient sanitation (toilets) facilities • No designated areas and dumpsters/bins for refuse dumping Public Policy-level barriers • Lack of funding and resource allocation constraints

4 Intrapersonal-level barriers

4.1 Limited knowledge, attitude and practice (KAP) of safe WASH practices and their negative consequences

Limited knowledge, attitude and practice (KAP) of safe WASH practices were identified as barriers to accessing safe WASH. Respondents reported that limited WASH education and awareness contributed to poor hygiene practices and inadequate water handling. These findings are based on qualitative data collected through interviews, where

probing questions were used to explore participants' understanding of WASH practices and their implications. Data collection continued until saturation was reached, ensuring that no new themes emerged.

"So, in Okpoko most people don't know how to maintain proper hygiene and good way of handling water". (IDI-12, State-level Policymaker)

In addition, the attitude and practice of the residents towards public health sanitation and hygiene were described to be poor. These negative attitudes were reported to manifest in the form of unacceptable cultural and social norms including indiscriminate or open dumping of refuse, open defecation and poor water storage systems. Respondents noted that these poor sanitary and hygiene practices were associated with frequent cases of diarrhoea, skin rashes, body odour, and genital infections that predominate in their community. While the findings primarily highlight cultural and social norms as drivers of these behaviours, the study did not explore in detail whether the availability of adequate facilities, such as functional toilets or waste disposal systems, may have also contributed to these practices.

"Some people don't want to go inside the toilet to defecate, they want it to be outside. They said that when you are defecating, that fresh air will be passing your anus and you will be enjoying, and then they are happy like that ... they defecate inside nylon and throw it inside the gutter and when it rains, it will carry those faeces and other wastes to our roads and pollute our environment." (IDI-15, Health Provider).

"Okpoko communities have this behavioural problem of dumping refuse indiscriminately, especially by the roadside and gutters....". (IDI-16, Community-level Policymaker)

"Some of those borehole owners do not clean their tank to the extent that they leave their tank open, and rats will enter it and people will buy it and drink". (FGD-04 Adolescent Group)

4.2 Limited awareness and understanding of WASH benefits

Some respondents demonstrated a limited understanding of the broader benefits of improved WASH. This was reflected in behaviours such as indifference toward hygiene and environmental sanitation. They perceived that poor WASH services can only affect children under five years of age, girls, and women, and did not fully recognize the potential health benefits for adults.

"The people living here act indifferently in paying attention to their hygiene and environmental sanitation. If you see where they cook and how dirty the environment is, you will be shocked, but they are not bothered". (IDI-19, LGA-level Policymaker)

4.3 Hostility of the inhabitants

It was noted that the residents of the study area were perceived as hostile. This attitude scares service providers; garbage collectors, health personnel, and implementing partners, from working in the community. Respondents attributed this hostility to factors such as security challenges in the area, high population density, and a perceived expectation of financial compensation before services could be provided. This was reported

to have impacted efforts to implement programs and services aimed at addressing poor access to WASH.

“Okpoko community is hostile; they are after settling them before you do anything for them even when they know that you came to assist them. Sometimes it baffles me why I should come to help someone and the person is asking me to pay before I should help; it’s funny. This draws people back from assisting them. There was a time UNICEF member that came there was attacked, maybe because of security challenges around the area and it is also highly populated.” (IDI-20, State-level Program Manager).

4.4 Interpersonal-level barriers

4.4.1 Lack of demand creation for improved sanitation and hygiene

There is a lack of demand for proper WASH facilities and services, particularly sanitation and hygiene reported to be significant barriers to accessing these essential services. Some respondents stated that it is a result of insufficient education on sanitation and hygiene practices, lack of access to information on safe WASH practices and low priority attached to WASH by the community members.

“... You know one problem is that we emphasize so much on water but do not have such a level of emphasis on sanitation and hygiene and hygiene is even more worrisome because in some schools we visited simple hygiene was not maintained....” (IDI-12, State Policymaker)

4.4.2 Refusal/lack of funds to pay sanitation levy

Another identified demand-side barrier was the non-payment of monthly refuse or sewage collection charges by most households hindering access to WASH services. The service charges cover the costs to remove refuse or thrash and are expected to be paid per household in the community. Respondents reported that while some people pay others are either not willing to pay or decline to pay for certain reasons. The reasons for the refusal or non-payment of such levy include conflicting political propaganda that the government had paid the levy on behalf of all the community members, limited financial resources, competing priorities for household expenses, dissatisfaction with previous sanitation services and inequitable sharing of sanitation costs.

“Last year there was a crisis in this regard because some political groups vying for the office of the President General in Okpoko campaigned against the payment of the sanitation levy by telling the community members that the state government have paid for the sanitation levy while it is false.... So political tussle affected immensely on how people respond to the issue of sanitation.” (IDI-17, Community-level Policymaker)

“Some household evacuates their faeces directly to the gutter because they cannot afford the high cost of evacuating soak away”. (IDI-05, Health Provider)

4.4.3 Unaffordability of water in the locality

Another issue is the inability to pay for water by some households due to the high cost. The respondents expressed concern about the high cost of purchasing water from the

only available source of water supply; the private boreholes which are essentially oriented to profit. It was noted that access to water is limited as households or individuals who cannot afford water costs resort to open defecation and poor hygiene practices.

“The toilet is very dirty and due to the high cost of water; people don’t have water to flush it and also to bathe. people will have body odour because they can’t shower or flush toilets”. (FGD-01, Adolescent Group)

4.5 Institutional level barriers

4.5.1 Poor multisectoral collaboration in WASH

WASH is an integrated approach that brings together multiple sectors to ensure increased efficiency, better outcomes, and sustainability. However, the study found that the required multi-sectoral collaboration was inadequate and lacked inclusivity. Some of the relevant sectors were reported to be working in silos and did not encourage an integrated approach to actions towards improving access to WASH not only in the study area but the state at large.

“In the state WASH program, sectoral collaboration or partnership is not okay.” There are other ministries like housing, infrastructure and even the private sector that should be there but they are not there. Sensitization and calling of stakeholders meeting to educate them that WASH is important.” (IDI-21, State-level Program Manager).

4.5.2 Inadequate inspection of WASH facilities by the relevant authorities

The lack of regular inspection of WASH facilities, particularly toilets, by the relevant LGA authorities (WASH Unit) was reported to contribute to non-compliance with proper sanitation practices, including water hygiene, in the slums. While households are responsible for the provision and maintenance of their own WASH facilities, the role of regulators in ensuring compliance through inspections and enforcement was noted to be lacking. Some community members noted that there used to be such inspections in the past, unlike the present years.

“Had it been that environmental health personnel from the LGA were coming, every house would have toilets and keep it very neat. You can imagine, those who have pit toilets, you find out that those pit toilets are not covered..... “In the past year 2015, only the Environmental Health Officers were involved in the management and provision of access to WASH. But now they are not doing anything, there is no form of inspection in this community”. (FGD-02, Women group).

4.5.3 Lack of maintenance and repair of existing public boreholes

The public boreholes which were supposed to serve as a major source of water were reported not to be functional due to lack of maintenance. As a result, the only available source of water is private boreholes, which respondents described as inadequate to meet the needs of the population. Community members perceived these private boreholes as ‘shallow holes’ and reported that the water from them was of poor quality, with some describing it as unfit for drinking or even bathing due to its smell and other physical characteristics. However, the study did not undertake a formal water quality analysis to scientifically assess these claims.

“Some of these boreholes are government-owned therefore, they are not working, no maintenance. They don’t fix it when it gets bad. It’s only the private borehole that functions properly.....but people can hardly get good water from these private boreholes because the boreholes are not deep. So, it doesn’t serve the people. You cannot drink it. And some of them, we can’t even bathe with it. The water smells...” (IDI-05, Health Provider)

4.5.4 Inadequate implementation and enforcement of rules against open defecation and indiscriminate refuse dumping

Although public rules against open defecation and indiscriminate dumping of refuse exist, most respondents reported that these rules are either not implemented or poorly enforced. The national strategy for promoting open defecation-free (ODF) communities includes steps such as segmentation into Ward Areas (WAs), promotion, follow-up mentorship, and progress monitoring to identify barriers and promoters. However, it was not evident from the findings that these steps were followed in the study area. Respondents noted that the lack of enforcement mechanisms and follow-up actions contributed to continued poor sanitation practices among the residents.

“Yes, the policy is there but implementation is the problem. We have sanitation rules but the issue is the enforcement.” (IDI-21, State-level Program Manager)

4.5.5 Inadequate waste management services/ facilities

Solid waste management in Okpoko is done by a third-party contractor who is responsible for collecting the refuse from all household doorsteps twice a week. However, this is not usually the case. Many of the community members reported irregular evacuation of refuse by the public service providers contracted to evacuate wastes. According to them, the service providers come once in one or two weeks. They noted that the irregularity in their services resulted in the accumulation of waste causing unsanitary conditions, odour, and breeding of rodents as the wastes stay longer within the confines of the households. This could lead to various health conditions such as malaria, diarrhoea, etc. In addition, the communities lack sufficient waste collection infrastructure, including inadequate waste collection facilities such as receptacles, dumpsters etc. leading to overflowing waste, unsanitary conditions and environmental health concerns.

“The ASWAMA people no longer come.... Like two weeks ago I brought out refuse to dispose, it is still there in front of my house because I do not know where to dispose of it” (FGD-01, Adolescent Group)

“There is no place and no container for refuse in this community, the government should look into that. People dig shallow holes to dump refuse and when it rains, it comes out to the surface. They should provide big waste bins for us.” (FGD, Men Group 1)

4.6 Community level barriers

4.6.1 Weak community leadership and governance

Another key supply-side barrier was the weak functionality of community-level leadership structures, such as WASHCOMs (Water, Sanitation, and Hygiene Committees) or WCAs (Water Consumers Associations) which are expected to support the provision

and maintenance of WASH facilities and services. In addition, the public boreholes were reported to be non-functional due to inadequate maintenance and vandalism, largely attributed to ineffective community leadership, which failed to safeguard and protect the boreholes, resulting in insufficient water supply in the communities. The community members expect that their leader should identify the issues of access to WASH and mobilize resources including funds and manpower to implement actions that address the issues. However, the leaders were said not to be responsive to the community's WASH plights.

"The majority of the people in the community do not pay attention to the sanitation exercise because our leaders are not checking what people do here and nobody is forcing people to do the right thing. There are no community rules and regulations against unsafe WASH practices". (FGD-04 Adolescent Group)

"Our community leaders are not doing anything to protect the borehole that the past governments did for us. No maintenance, all the parts have been stolen and they are just there. All their interest is no sanitation levy". (FGD-04 Adolescent Group)

However, some community leaders stated that they are doing their best to ensure safe WASH but that their people are not complying and are difficult to rule or coordinate. A community leader cited an example of how the community members always fought private service providers (who are helping to evacuate waste) and the community task force on sanitation which resulted in them withdrawing their services amidst the crises.

"... last month February this year 2024. The community task force on sanitation went to collect the sanitation levy, but some compounds refused to pay the sanitation levy and fought the task force which resulted in serious injuries and even death. So, the community was in serious disarray, some community security personnel ran away from the community looking for their safety elsewhere due to the crisis". (IDI-16, Community-level Policymaker)

4.6.2 Inadequate water supply system to serve the population

The rapid urbanization resulted in a surge in population growth of Okpoko from 31,000 in 1978 to 128,417 in 1998, to 177,608 in 2007 and 670,000 in 2022 [24, 25] putting pressure on only two available boreholes (a major source of water in the study areas). The rapid population growth has led to the insufficient water supply that meets the needs of the growing population including irregular or unavailability of water in the study communities. Hence, water supply becomes a major barrier to access to safe WASH for all who live in Okpoko.

"Our (Okpoko community) population is massive so the number of boreholes here cannot serve us". (IDI-18, Community-level Policymaker).

4.6.3 No designated areas and bins for refuse dumping

There is a lack of designated waste disposal stations and bins for dumping refuse in the communities. This facilitated the practice of open dumping of refuse along walkways and inside gutters. This results in littering, unpleasant odours and aesthetics and an increased risk of spread of WASH-borne or related diseases.

“There is no marked dumping ground so they dispose of waste anywhere, people come out at night to dump waste in the gutter....” (FGD-05, Men group).

4.6.4 Absence of and inadequate toilet facilities

The respondents reported that most of the buildings/households lacked access to adequate toilet facilities. These include an insufficient number of toilets (considering the population of people living in each household), poor maintenance and cleanliness, inadequate water supply for the toilet system, and no toilet handwashing facilities. The situation was reported to be the major cause of open defecation making the environment unpleasant.

“Many don’t have good toilets and because some are always dirtied some people will choose to go where they have good public toilets and here is overpopulated, some people will wait for their turn to use the toilet in their public yards and decide to use nylon to defecate and pour it in the drainage.” (IDI-08, Community-level Policy-maker)

4.7 Public policy-level barriers

4.7.1 Funding constraints

Poor budgetary allocation and no release of funds for WASH activities or services at state and LGA levels were found to be a major barrier to the provision of WASH services. Respondents reported that few programs implemented at the state level are donor-supported/funded which usually have short life spans. According to State and LGA-level program implementers, funding constraints negatively affect service delivery. This leads to the non-implementation of WASH programs, non-inspection of WASH facilities, and lack of monitoring and evaluation of WASH services particularly in semi-urban including Okpoko communities and flood-prone areas where there is obvious poor access to WASH. While respondents highlighted the lack of budgetary allocation and fund release as key issues, the study did not conduct a detailed analysis of WASH budgetary provisions, the percentage of funds released, or their utilization and this has been identified as one of the gaps of this study and area for future research.

“There has never been allocation to WASH over the years. For instance, last year (2023) we had a budget for WASH but we didn’t get approval.....Yes for release, it is zero Naira, the budget is One hundred million Naira” for this year and last year was Three hundred million Naira.” (IDI-20, State-level Program Manager)

Community members validated the report of non-inspection of the WASH facility (toilet) in the study communities by any authority over ten years ago.

“Some yards do not have toilets...some toilets are dirty and full, but nobody is checking them..... the people that have this job are the health personnel from environmental health. I remember when I was small, they used to come for inspection and if you do not have the necessary sanitation and hygiene facilities, they will sue you. But they don’t do it again.....” (FGD-02, Women Group)

5 Discussion

The study findings showed that several demand and supply-side barriers, cutting across all the levels of the socio-economic model were contributing to poor access to WASH services in urban slums. It also highlights the interplay of how the barriers at various levels limit access to WASH underscoring the critical need for integrated approaches to addressing both demand and supply sides of the access equation.

The intrapersonal level barriers play a significant role in limiting access to safe WASH practices in the slums. Specifically, poor knowledge, attitude and practice on safe WASH and its benefits, lack of demand creation for improved WASH, refusal to pay sanitation levies, and hostility towards WASH promoters hinder access to safe WASH services. The findings imply a huge gap in knowledge and understanding of the importance of safe WASH particularly in disease prevention. The residents' hostility towards WASH promoters could stem from perceived interference in traditional practices and mistrust of external interventions. This is not surprising as slums are usually inhabited by individuals from varied backgrounds with diverse beliefs and values. These beliefs and values shape their ability to appreciate and take up certain health promotion activities or interventions. To address these behavioural barriers, effective and inclusive health promotion strategies including behavioural change communication using appropriate information, education and communication (IEC) materials should be employed to engage and educate community members on WASH and the benefits of safe WASH practices.

Consequently, the interpersonal level barriers show that there is low demand creation for improved or safe WASH practices whereas the perceived low risk of WASH-borne diseases may be attributable to poor practices. The critical challenge of inadequate demand for safe WASH practices and refusal to pay sanitation levy despite its importance demonstrated low appreciation of the value of safe WASH practices. This could be attributed to limited awareness of WASH and its benefits and prioritization of other basic needs over WASH. Our findings are consistent with previous studies that reported limited awareness of WASH benefits to be linked to demand-side barriers [26, 27].

The significant role of household financial constraints, which is also an interpersonal level barrier in limiting access to safe WASH in the slums is also highlighted in the study. These findings align with previous research demonstrating the negative effect of financial constraints on WASH access in low-income settings [28]. The economic burden of WASH expenses, coupled with competing priorities for limited household resources, has been previously reported to exacerbate this challenge [29].

At the institutional level, the study highlights the critical role of service delivery barriers in limiting access to WASH services in the slums. Specifically, the inadequate and irregular evacuation of waste hampers access to WASH. More so, it will contribute to unsanitary conditions and increase the risk of WASH-borne and WASH-related diseases. The findings align with previous studies demonstrating that inadequate WASH services contribute to increased morbidity and mortality rates in urban slums [1, 10–13, 15, 16, 19].

The findings show the significant effects of institutional-level barriers in limiting access to WASH in the urban slums. These barriers perpetuate inequalities and exacerbate health risks for the residents. Addressing these institutional and policy-related barriers requires strengthening multisectoral collaboration and coordination, capacity-building

of stakeholders, community engagement and participation in the delivery of WASH services in urban slums.

The community-level barrier findings that inadequate water supply systems, insufficient sanitation facilities including insufficient toilets and non-provision of solid waste management facilities are infrastructural barriers limiting the availability and quality of WASH services and facilities in the slums are similar to previous research evidence on WASH [6–7]. It was also inferred that the lack of safe and adequate sanitation facilities, coupled with poor waste disposal practices, perpetuates the persistent open defecation and indiscriminate dumping of refuse in the slums. This emphasizes the need for urban planning prioritizing and investing in WASH infrastructure development in the slums. The study results have important implications for policymakers and implementers.

At the public policy level, the non-budgetary allocation and non-released funds for WASH found in this study indicate low prioritization of WASH services at both state and LGA levels and low political will to address such urban slum challenges. This has been reported by a study in Kenya which found non-provision of WASH services and facilities due to lack of funds as a result of insufficient prioritization of WASH in the National development agenda [19]. The lack of funds at both state and LGA levels may be attributable to the poor WASH service delivery and some infrastructural-related barriers identified in the study. Our finding is similar to a previous study on economic and financial analysis of water and sanitation services in urban slums of India that reported insufficient funding as a barrier to the provision of WASH services and facilities [28]. Our findings are similar to a study conducted in Southern Africa that found access to WASH to be hindered by inadequate financing, ineffective local community engagement in WASH interventions and low investments in WASH infrastructure [30].

To achieve SDG goal 6 in urban slums, effort should be made to develop and implement context-specific solutions such as integrating WASH programming simultaneously addressing supply and demand-side barriers, behaviour change communication - implementing awareness campaigns and Community-Led Total Sanitation (CLTS) initiatives and investments prioritizing water supply, sanitation, and hygiene infrastructure.

6 Strength and limitation

A major strength of the study is the in-depth exploration of perspectives of different categories of stakeholders (policy-makers and program implementers from diverse sectors and levels and community members) who play various roles in the provision of WASH services revealing barriers that are consistent with literature. In addition, the qualitative methods employed in the study offered in-depth insights into WASH access challenges. However, it is important to acknowledge that this study has some limitations. The findings may not be generalizable to other regions due to the specific context in which the research was conducted. While this study provides valuable insights into the experiences, perceptions, and practices of community members and stakeholders regarding WASH issues, we acknowledge that a quantitative approach could provide additional insights and enhance our understanding of the WASH situation. A future study using a quantitative approach would be warranted to collect data on key indicators such as access to improved water sources, sanitation coverage, and hygiene practices. This would provide a more comprehensive picture of the WASH situation and inform policy and programming decisions. Specifically, quantitative data could help to identify

the prevalence of WASH-related health issues and track changes in WASH coverage over time. By doing so, researchers and policymakers could gain a more nuanced understanding of the WASH situation and develop more effective strategies for improvement. There was also a lack of further exploration and analysis into some of the reports of the respondents such as a lack of budget impact analysis which is an identified gap and area for future research.

7 Conclusion and policy recommendation

The study highlights the complex and multi-faceted barriers affecting access to and provision of WASH services in urban slums. The findings reveal a range of demand-side and supply-side barriers, spanning from individual knowledge and attitudes to institutional and policy-level constraints. These barriers perpetuate the inadequate provision of WASH services, exacerbating health risks and undermining the well-being of urban slum residents.

By addressing these barriers and implementing the policy recommendations below, governments, NGOs, and other stakeholders can work towards improving access to and provision of WASH services in urban slums, ultimately enhancing the health, well-being, and quality of life for urban slum residents and ultimately, contributing to the achievement of SDG goal 6.1 and 6.2.

To address the pressing WASH challenges in the urban slums and improve the health and well-being of the residents, the following recommendations are crucial;

- Integrate WASH education into existing community programs to enhance knowledge, attitudes, and practices among urban slum residents.
- Strengthen multisectoral collaboration among government agencies, NGOs, and private sector stakeholders to improve WASH service delivery and address institutional barriers.
- Enhance institutional capacity for WASH inspection and enforcement by providing training and resources to strengthen the capacity of local authorities to inspect WASH facilities and enforce standards.
- Implement CLTS initiatives or programs that will empower communities to take ownership of sanitation improvements and promote behaviour change.
- Establish an accountability mechanism to ensure community leaders are responsible for WASH service delivery and maintenance.
- Increase funding and resource allocation for WASH infrastructure at State and LGA levels for acquiring and maintaining WASH infrastructure including water supply systems, sanitation facilities, and waste management services.

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Author contributions

UE and OO conceived the idea and designed the study. UE, IO, CO and JN coordinated the data collection, transcription, and transformation. UE, IO, and CO performed data coding and analysis. UE drafted the manuscript. All authors contributed to the manuscripts and approved the submitted version.

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Data availability

Availability of Data and Materials The data used and/or analyzed during the current study are available from the corresponding author, Uchenna Ezenwaka upon reasonable request.

Code availability

Not applicable.

Declarations**Ethics approval and consent to participate**

Ethical approval was obtained from the Health Research and Ethics Committee of the University of Nigeria Teaching Hospital Ituku-Ozalla, Enugu Nigeria (UNTH/HREC/2023/10/639). This study was conducted following the Declaration of Helsinki. All participants provided both written and verbal informed consent. Participants were informed of the purpose of the research, the rights of participants and measures that would be taken to protect them and their data. Hence, participation was voluntary, and confidentiality was assured.

Competing interests

The authors declare no competing interests.

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